NIC26



Migrating from Hybrid Active Directory to Entra ID Cloud Native. A Technical Deep Dive

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Microsoft MVP Most Valuable Professional 2026-2027

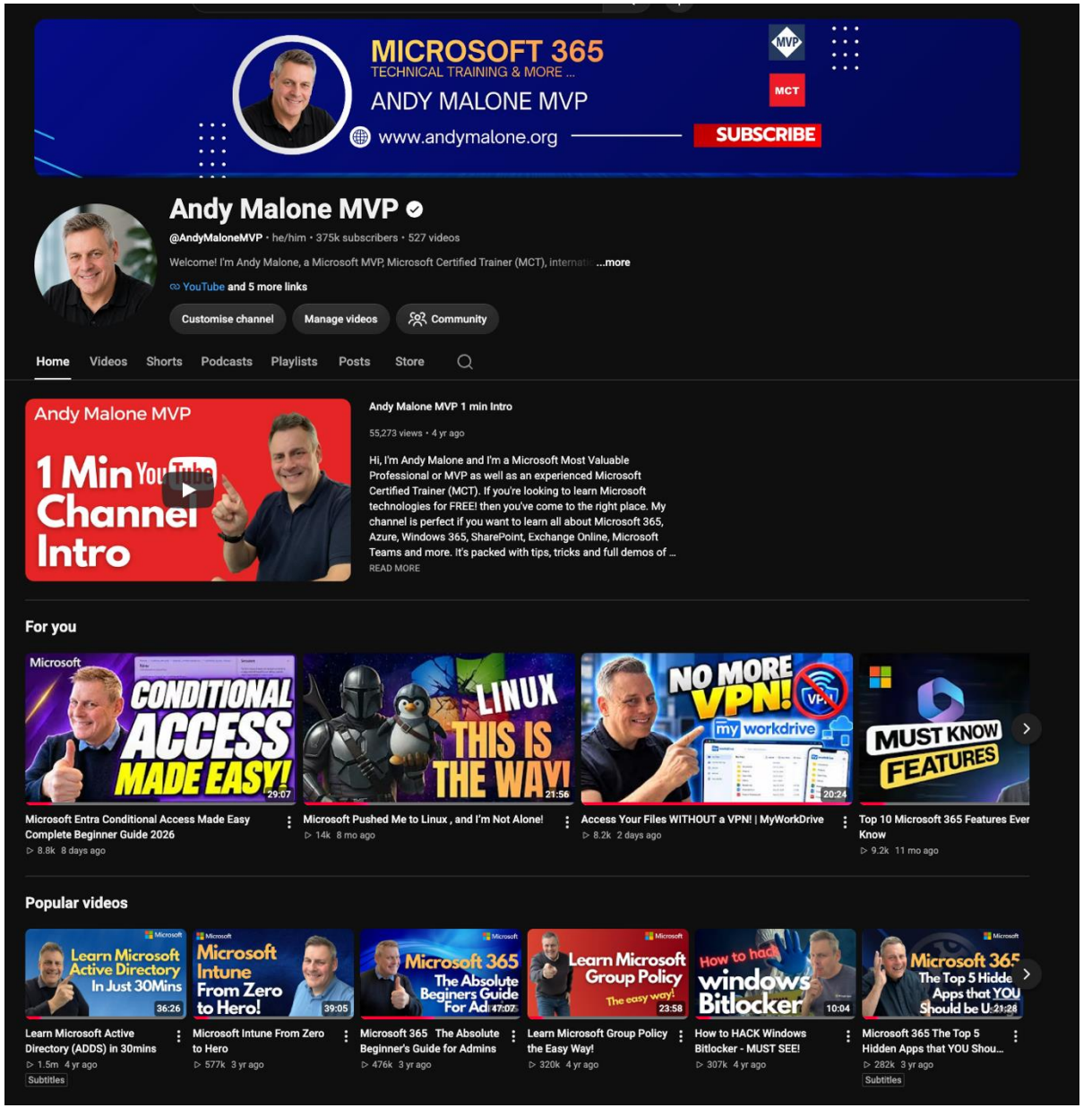
YouTube

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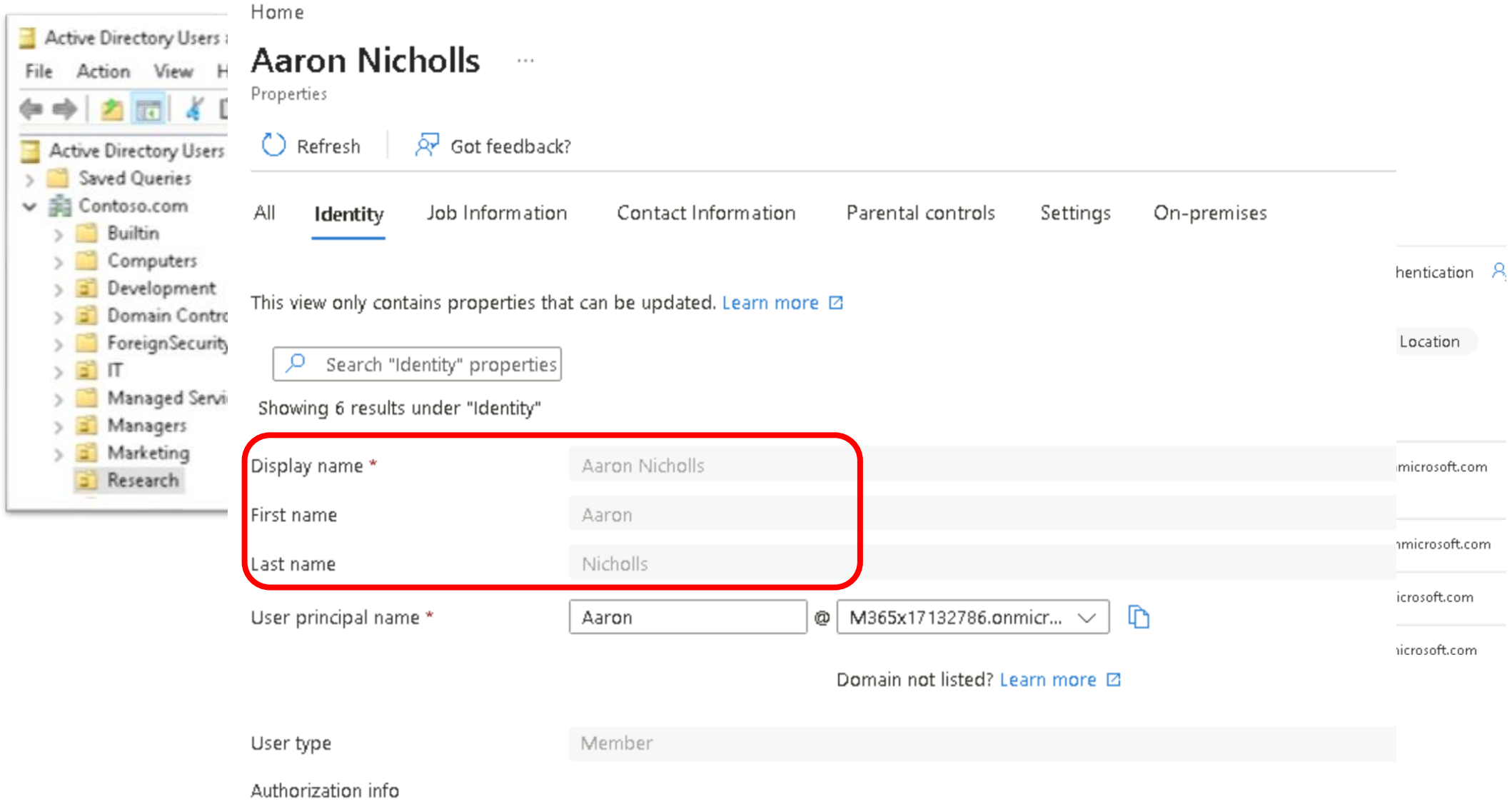
15 Million Views
376k Subs



Today’s Goal



The State of Hybrid in 2026



The State of Hybrid in 2026

Stage 1 Cloud Attached

Tenant exists, M365 in use

Stage 2 Hybrid (Most Common Config)

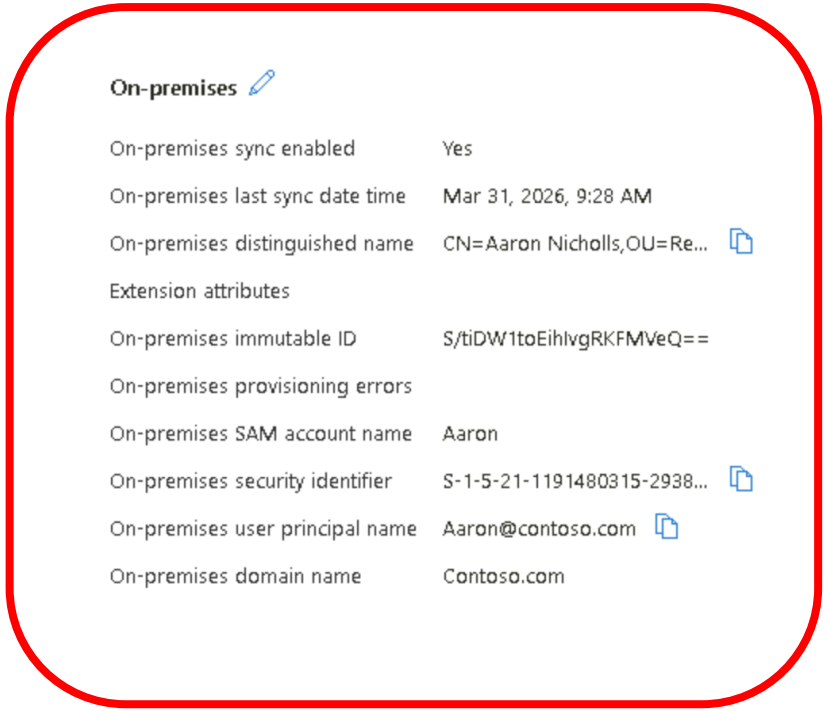
Sync AD → Entra ID

Stage 3 Cloud First

New objects born in cloud

Stage 4 Cloud Only

AD DS retired



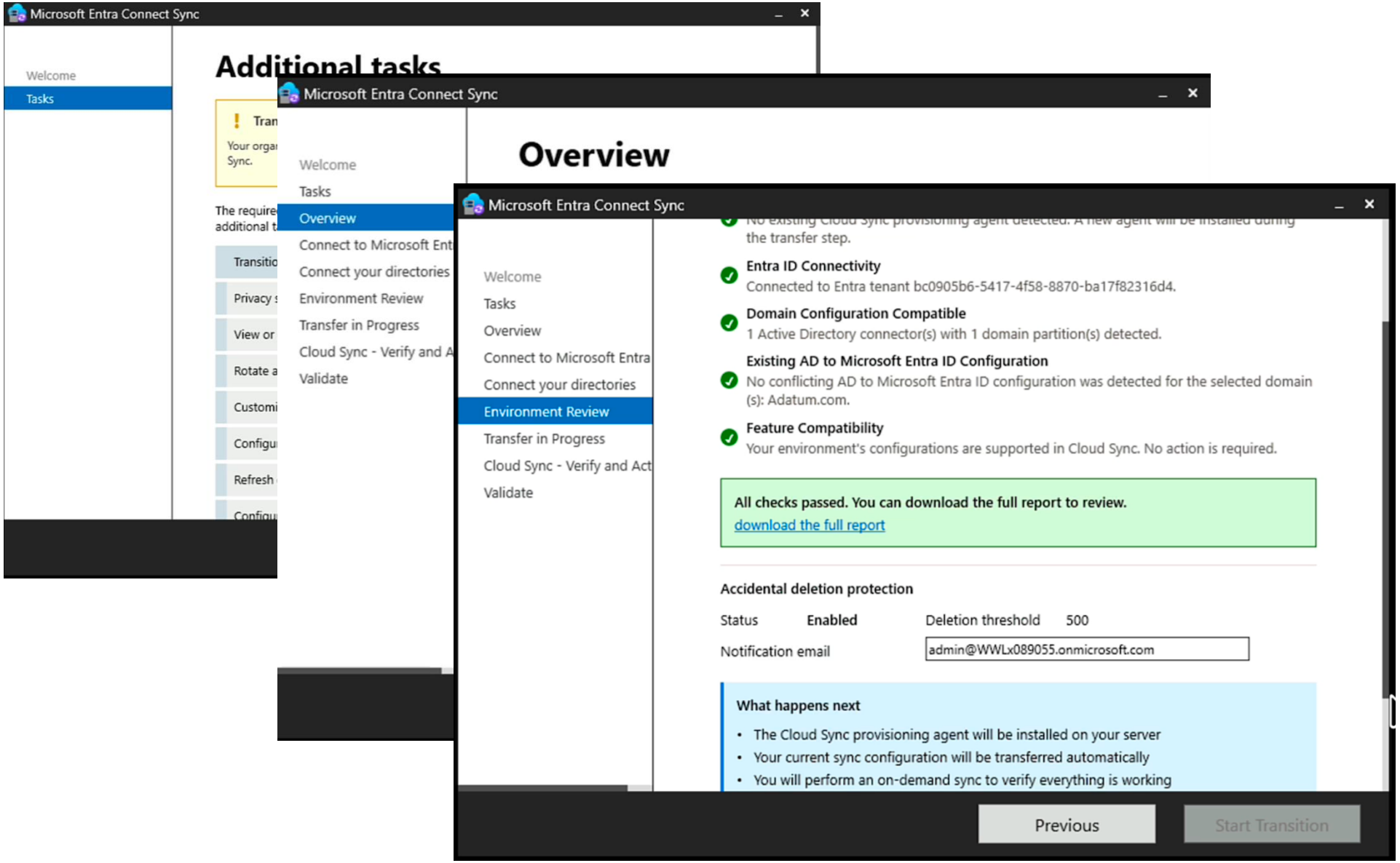


Entra ID Connect Sync

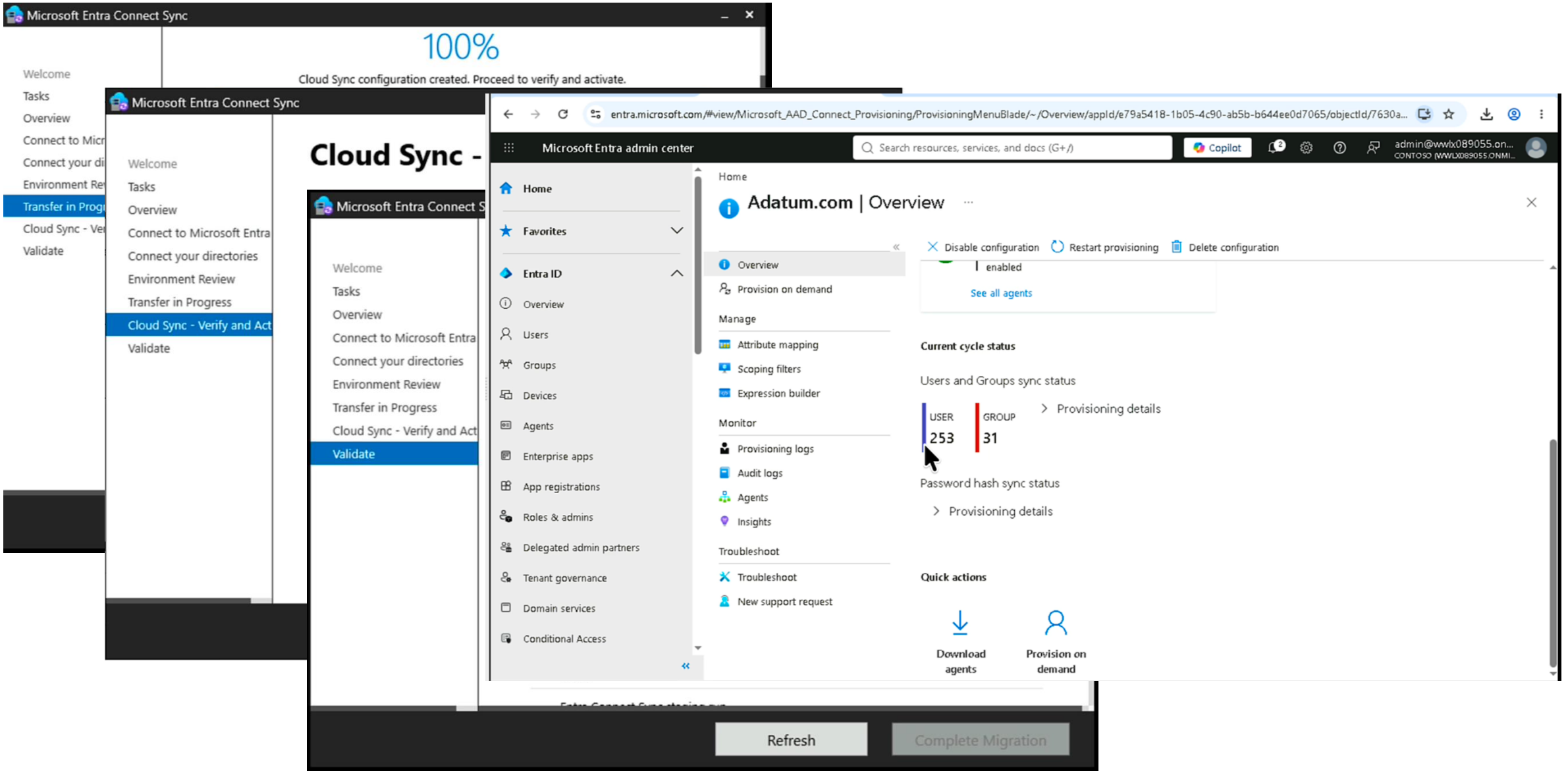
Walkthrough

Step 1 Migrating to CloudSync

One Small Step: Entra Connect Sync to Cloud Sync Migration



One Small Step: Entra Connect Sync to Cloud Sync Migration

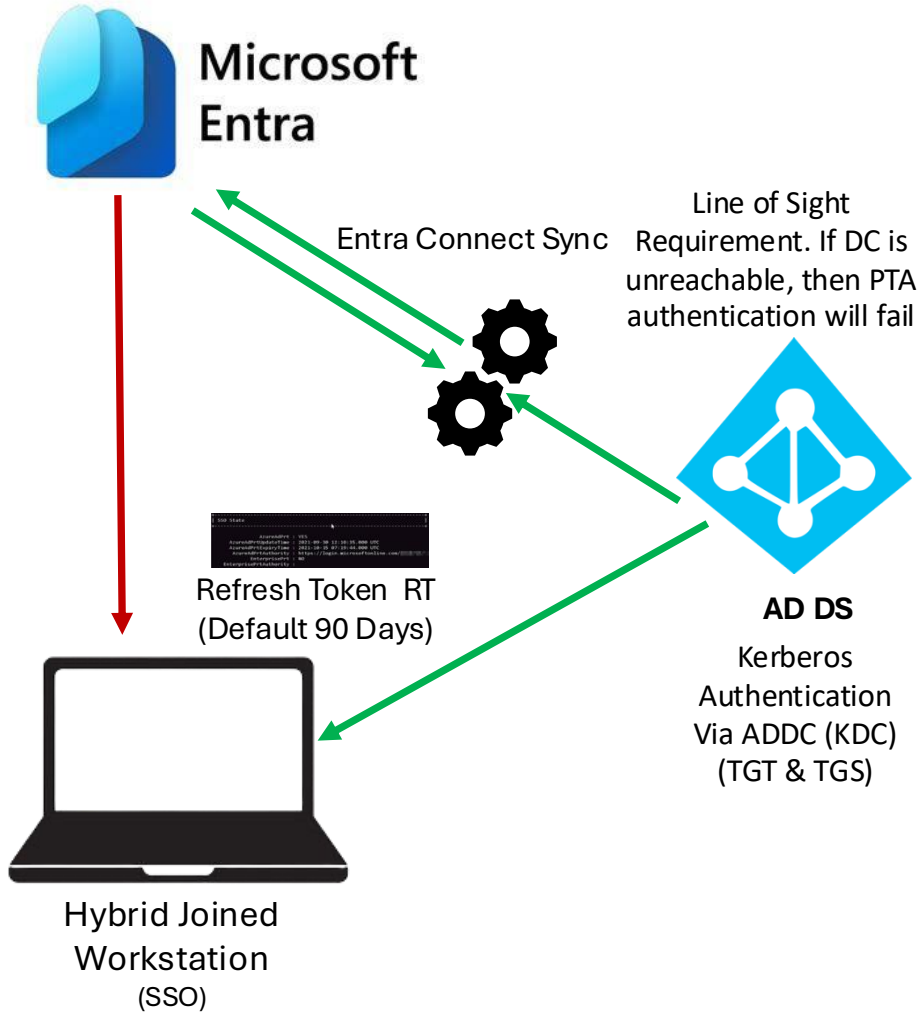




A Word about Device Source of Authority (SOA)

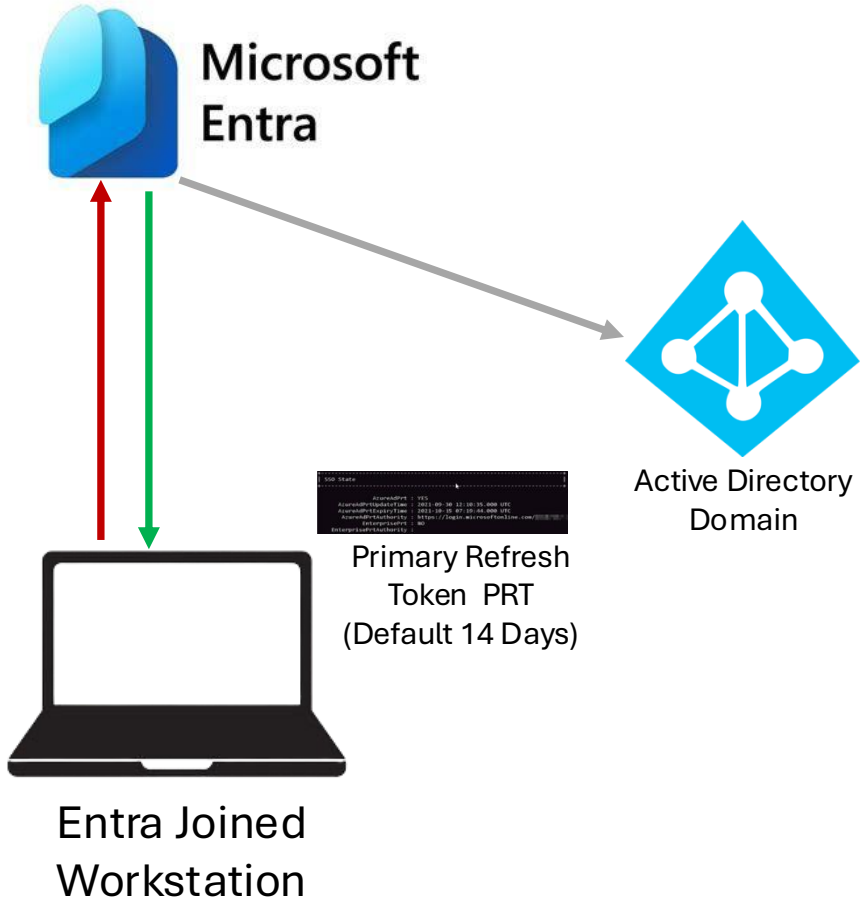
Device SOA - Hybrid Joined Device Authentiction

- For a **Hybrid Joined** workstation, identity originates from **on-premises Active Directory (AD)**.
- Device is domain-joined first, requiring **line-of-sight to a domain controller** for initial user logins, password changes, and Group Policy processing.
- While it registers with Entra ID to gain cloud SSO via RT, the **AD DC remains the source of authority**.
- Since a standard AD-joined device is **not** registered with Entra ID for device identity, thus it does **not receive a PRT.**
- Instead of a PRT, User receives a **Refresh Token** which is a standard Refresh Token is **typically bound to a single application or client**
- **Token Protection** for Conditional Access is now available to **Entra ID P1**
- This feature **cryptographically binds refresh tokens** to the specific **Windows, MacOS & iOS** devices on which they were issued, preventing stolen tokens from being replayed on unauthorized machines.



Device SOA - Entra Joined Device Authentiction

- For an **Entra Joined** workstation, identity originates in the cloud.
- The device authenticates directly to **Microsoft Entra ID** using internet connectivity.
- Users sign in with cloud credentials, and the device receives a **Primary Refresh Token (PRT)** for seamless single sign-on (SSO) to cloud resources.
- The PRT is cryptographically tied to each unique device, typically using a **TPM)** to prevent unauthorized extraction or replay attacks.
- It does not depend on on-premises infrastructure for login, allowing full functionality for remote users without VPN or domain controller (DC) line-of-sight.





Let the Great Migration Begin. Source of Authority SOA Internals

First! The Big Decision



SOA Migration (Prerequisites)

- Hybrid Administrator
- Application Administrator
- Cloud Application Requirements

Admin Roles



- For apps calling into the onPremisesSyncBehavior Microsoft Graph API, the User-OnPremisesSyncBehavior.ReadWrite.All permission scope needs

Permissions



- Minimum Licence
- Entra ID Free Licence

Licence Requirements



- Minimum version is 2.5.76.0. To use
- Contact SOA, version 2.5.79.0.

Connect Sync Client



Minimum version is 1.1.1370.0

Cloud Sync Client



Source of Authority (SOA) Explained!

SOA The system that owns an object's core attributes and lifecycle. Writes here are authoritative; writes elsewhere are either blocked or overwritten on the next sync cycle.

ON-PREM SOA (classic hybrid)

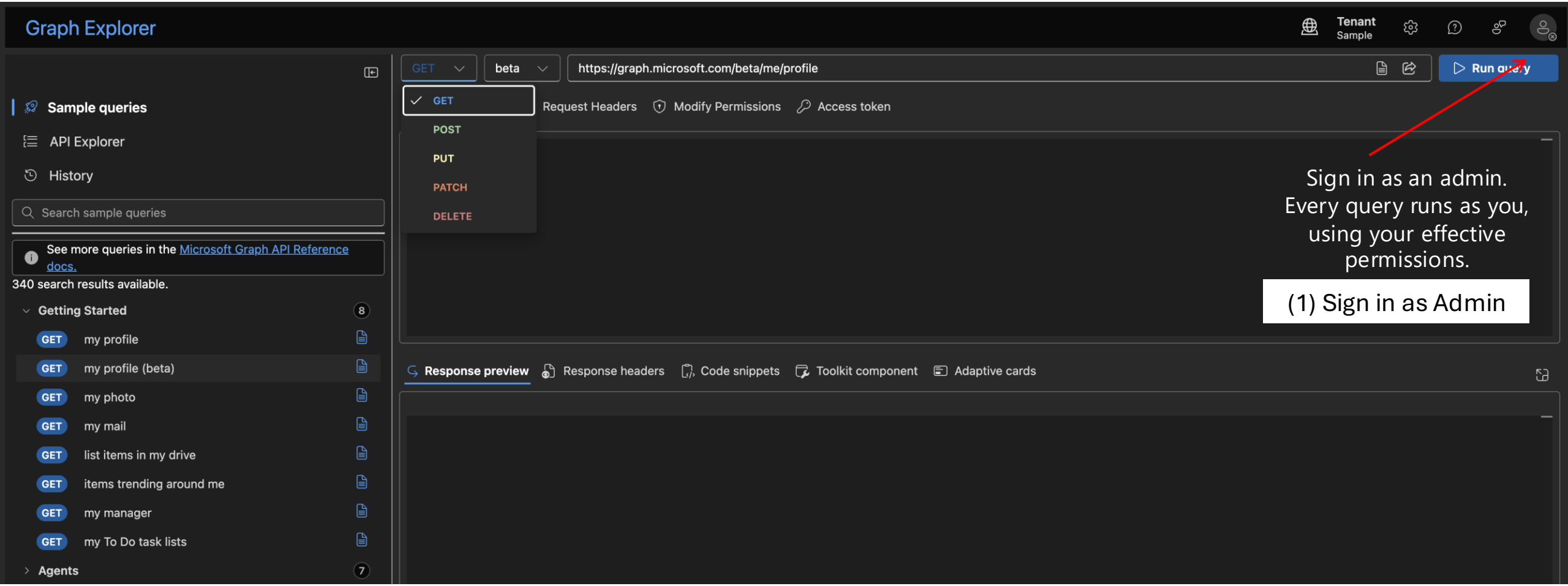
- AD DS mastered; properties flow
  - AD → Entra
- isCloudManaged = false on the Entra object
- Cloud edits are rejected or overwritten
- Account recovery, licensing touch both sides

CLOUD SOA

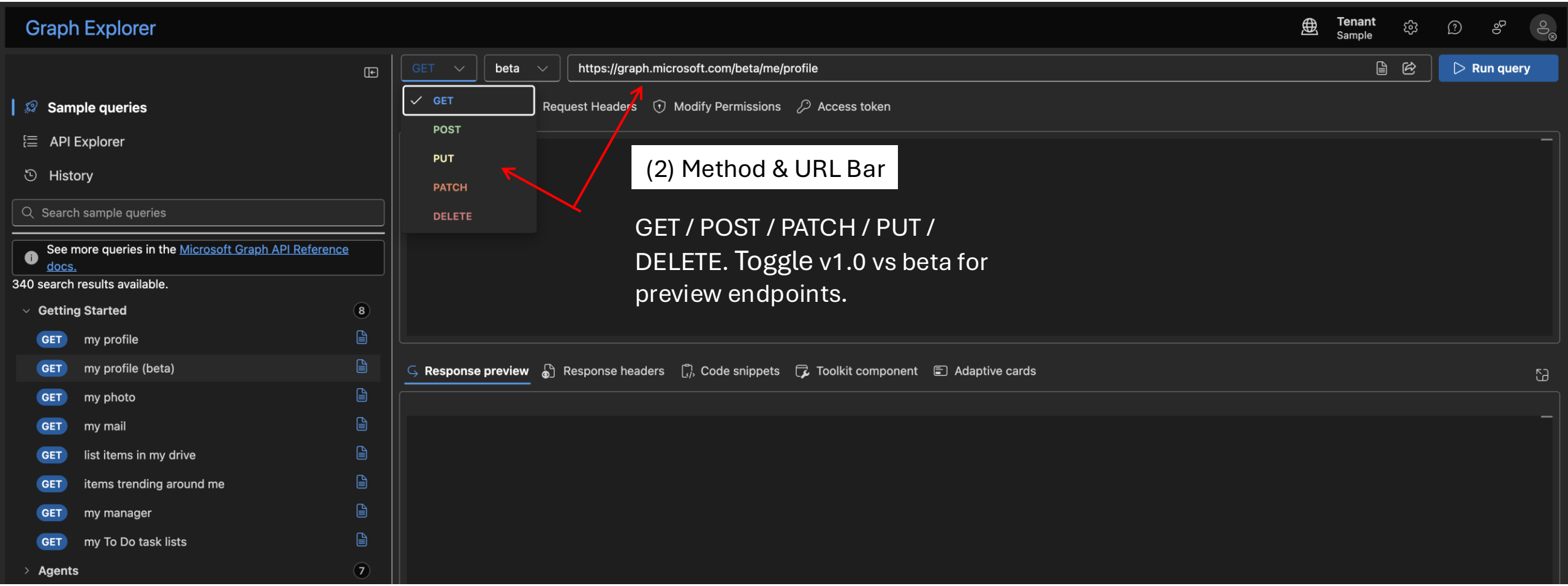
- Entra ID is authoritative for the object
- isCloudManaged = true (the flag we'll flip)
- AD changes are ignored (not reverse-synced)
- Unlocks Governance, Lifecycle Workflows, PIM depth



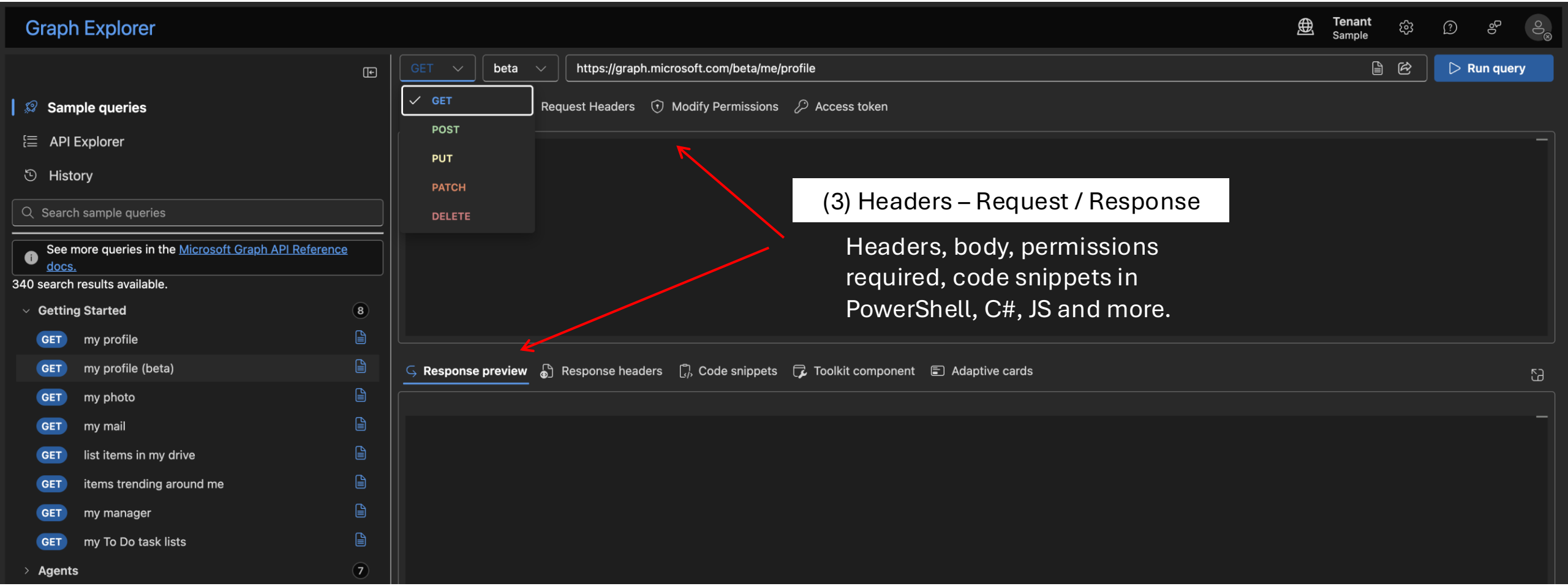
Graph Explorer
A Quick Guided Tour



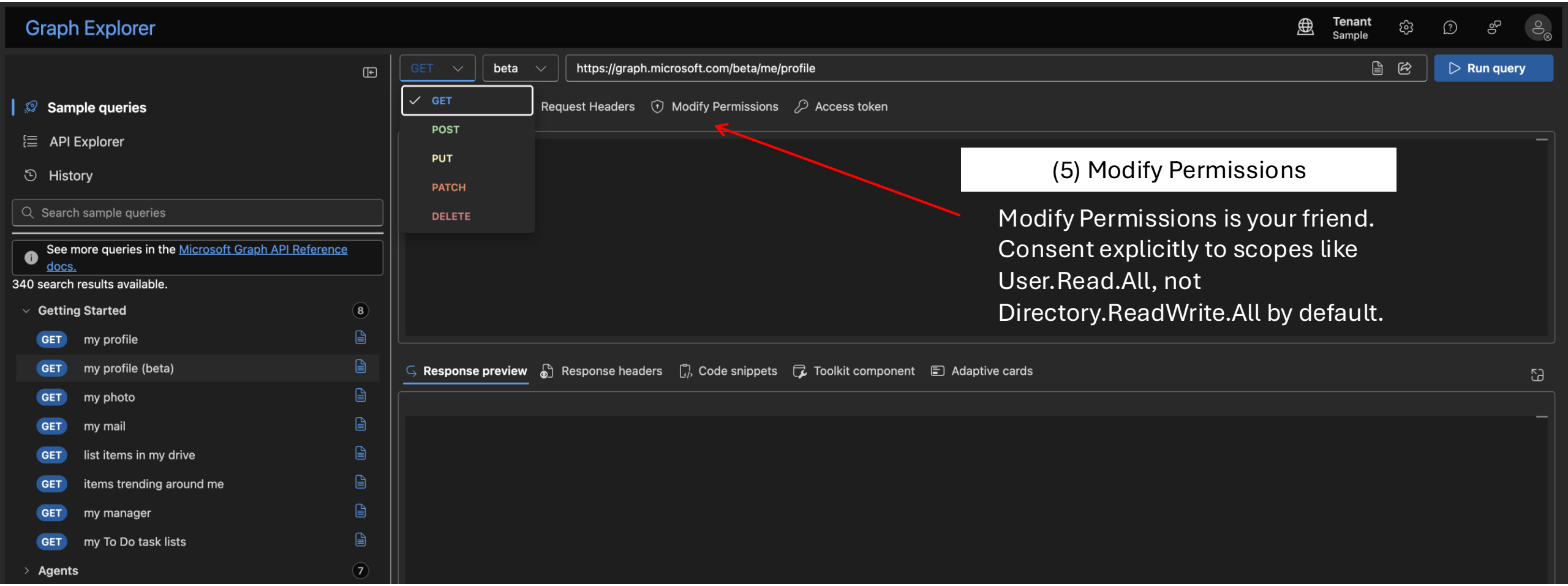
Graph Explorer



Graph Explorer



Graph Explorer



Graph Explorer

Graph Permissions: Use Principle of least-privilege

Delegated / Application Scopes	
`User.Read.All`	Enumerate users
`User.ReadWrite.All`	Edit user properties
`Group.Read.All`	Enumerate groups
`Group.ReadWrite.All`	Edit group properties
`User-OnPremisesSyncBehavior.ReadWrite.All`	Flip user SOA (the new one)
`Group-OnPremisesSyncBehavior.ReadWrite.All`	Flip group SOA
`Organization.ReadWrite.All`	Disable tenant-level sync
`Directory.AccessAsUser.All`	Delegate via admin session

Entra Directory Roles (built-in)	
Global Administrator	Full tenant — avoid in scripts
Hybrid Identity Administrator	Purpose-built for this work
User Administrator	User create / update / delete
Groups Administrator	Group CRUD + membership
Application Administrator	Grants required for SOA consent
Authentication Administrator	Password resets, MFA methods
Security Reader	Read Conditional Access, signins
Reports Reader	Audit & signin logs only



Best practice: dedicated app registration, assign Hybrid Identity Administrator via PIM with 4-hour activation, grant only the two SOA scopes.



Group Source of Authority (SOA)

Groups first, always. User SOA depends on this working cleanly.

Why the New SOA Patch Switch Changes the Game

BEFORE

Delete.
Re-create.
Restore.

Every unsync resulted in a new objectId. Licenses, group memberships, device relationships, conditional access targeting — all at risk.

NOW

One PATCH request.

```
PATCH /v1.0/users/{id}/onPremisesSyncBehavior
{ "isCloudManaged": true }
```

- objectId preserved · licenses retained
- Sync stops for that single object only
- Can be reversed (rollback supported)

Groups first — always



WHY GROUPS BEFORE USERS

- Group membership drives application access
- Users migrating while still member of on-prem group create dependency issues
- Group SOA is GA. User SOA builds on top of it
- Microsoft's stated sequence:
  groups → users → tenant

WHAT HAPPENS AT THE MOMENT OF SWITCH

- Entra Connect detects isCloudManaged = true
- Connect Sync stops syncing that group
- The group becomes editable in Entra ID
- You can add cloud-only users as members
- Mail-enabled groups stay in Exchange Online management
- Optional: provision group back to AD via Cloud Sync

Group Example Marketing



Scenario: 'Marketing' (Security Group) grants access to a SaaS app via Entra SSO. The group lives in OU=Finance in AD. We want to manage membership from the cloud so auditors can run access reviews.

```
# 1. Connect with least-privilege scopes
Connect-MgGraph -Scopes "Group.Read.All","Group-OnPremisesSyncBehavior.ReadWrite.All"

# 2. Find the group by display name
$g = Get-MgGroup -Filter "displayName eq ‘Marketing'" -Property Id,OnPremisesSyncEnabled
$g | Format-List Id,DisplayName,OnPremisesSyncEnabled
# OnPremisesSyncEnabled : True   (confirmed, safe to proceed)

# 3. Flip the SOA
$uri  = "https://graph.microsoft.com/v1.0/groups/$($g.Id)/onPremisesSyncBehavior"
$body = @{ isCloudManaged = $true } | ConvertTo-Json
Invoke-MgGraphRequest -Method PATCH -Uri $uri -Body $body

# 4. Verify — should read True
(Invoke-MgGraphRequest -Method GET -Uri $uri).isCloudManaged
```

Group SOA — Prerequisites & Gotchas



PREREQUISITES

- Entra Connect Sync build 2.5.76.0 or later
- Or Entra Cloud Sync agent build 1.1.1370.0+
- Group-OnPremisesSyncBehavior.ReadWrite.All
- Group.ReadWrite.All delegated / app perm
- Application Admin or Cloud App Admin role
- Group must already be syncing successfully

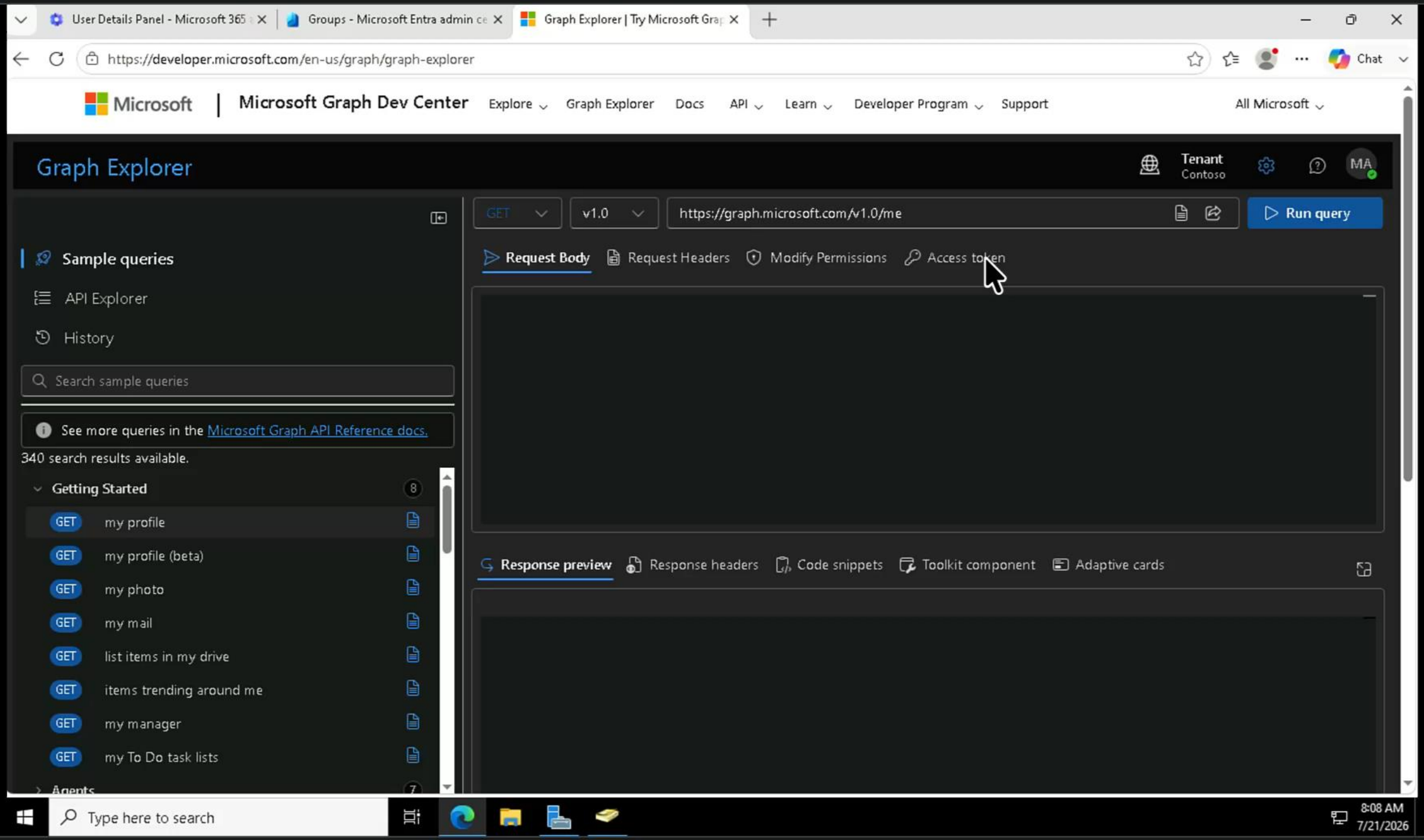
GOTCHAS

- Dynamic-group rules won't apply while on-prem synced
- Nested AD group memberships — flatten first
- Want group back in AD? Pre-create a DN extension attr
- Rollback: remove cloud-only members before reverting
- Mail-enabled groups! Manage via Exchange Online only
- Always pilot: 5-10 groups, full business-week soak



Demo

Group SOA Flipping

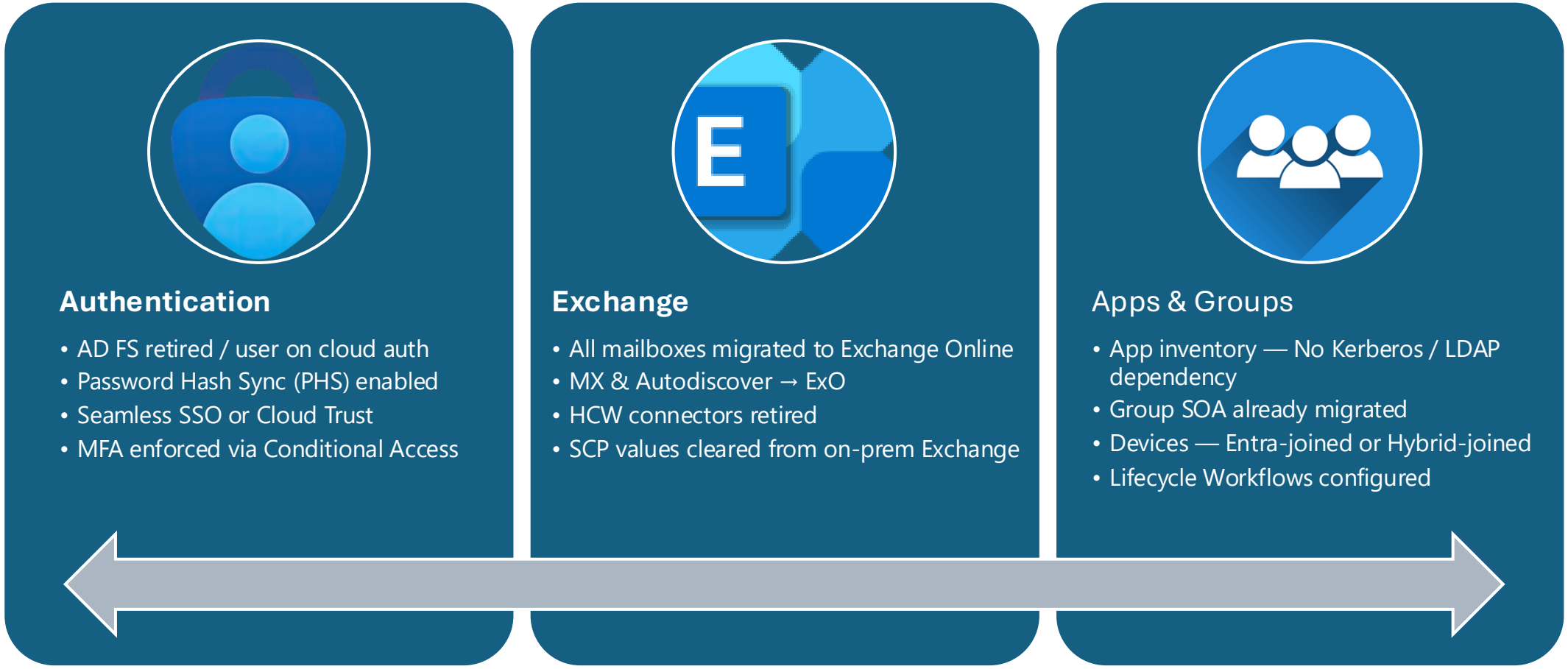

User Details Panel - Microsoft 365
Groups - Microsoft Entra admin ce
Graph Explorer | Try Microsoft Gra
https://developer.microsoft.com/en-us/graph/graph-explorer
Chat
Microsoft
Microsoft Graph Dev Center
Explore
Graph Explorer
Docs
API
Learn
Developer Program
Support
All Microsoft
Graph Explorer
Tenant
Contoso
MA
GET
v1.0
https://graph.microsoft.com/v1.0/me
Run query
Sample queries
API Explorer
History
Search sample queries
Request Body
Request Headers
Modify Permissions
Access token
See more queries in the Microsoft Graph API Reference docs.
340 search results available.
Getting Started
8
GET my profile
GET my profile (beta)
GET my photo
GET my mail
GET list items in my drive
GET items trending around me
GET my manager
GET my To Do task lists
Agents
7
Response preview
Response headers
Code snippets
Toolkit component
Adaptive cards
Type here to search
8:08 AM
7/21/2026



User Source of Authority (SOA)

Finally, the feature we've been asking for — But Gotchas still exist

User SOA. Preparation is Everything!



Rule of thumb: if a user can already work with the on-prem DC offline, they are SOA-ready.

Worked example — sara.davis@contoso.com

```
# Step 1: Connect with minimal scopes
Connect-MgGraph -Scopes "User.Read.All","User-OnPremisesSyncBehavior.ReadWrite.All"

# Step 2: Load the user & sanity-check state
$u = Get-MgUser -UserId "sara.davis@contoso.com" -Property
Id,DisplayName,OnPremisesSyncEnabled,OnPremisesImmutableId
"$($u.DisplayName)  Synced=$($u.OnPremisesSyncEnabled)  ImmutableId=$($u.OnPremisesImmutableId)"
# Sara Davis  Synced=True  ImmutableId=aUJk8LbYzk2vG4s2rCx1Yg==

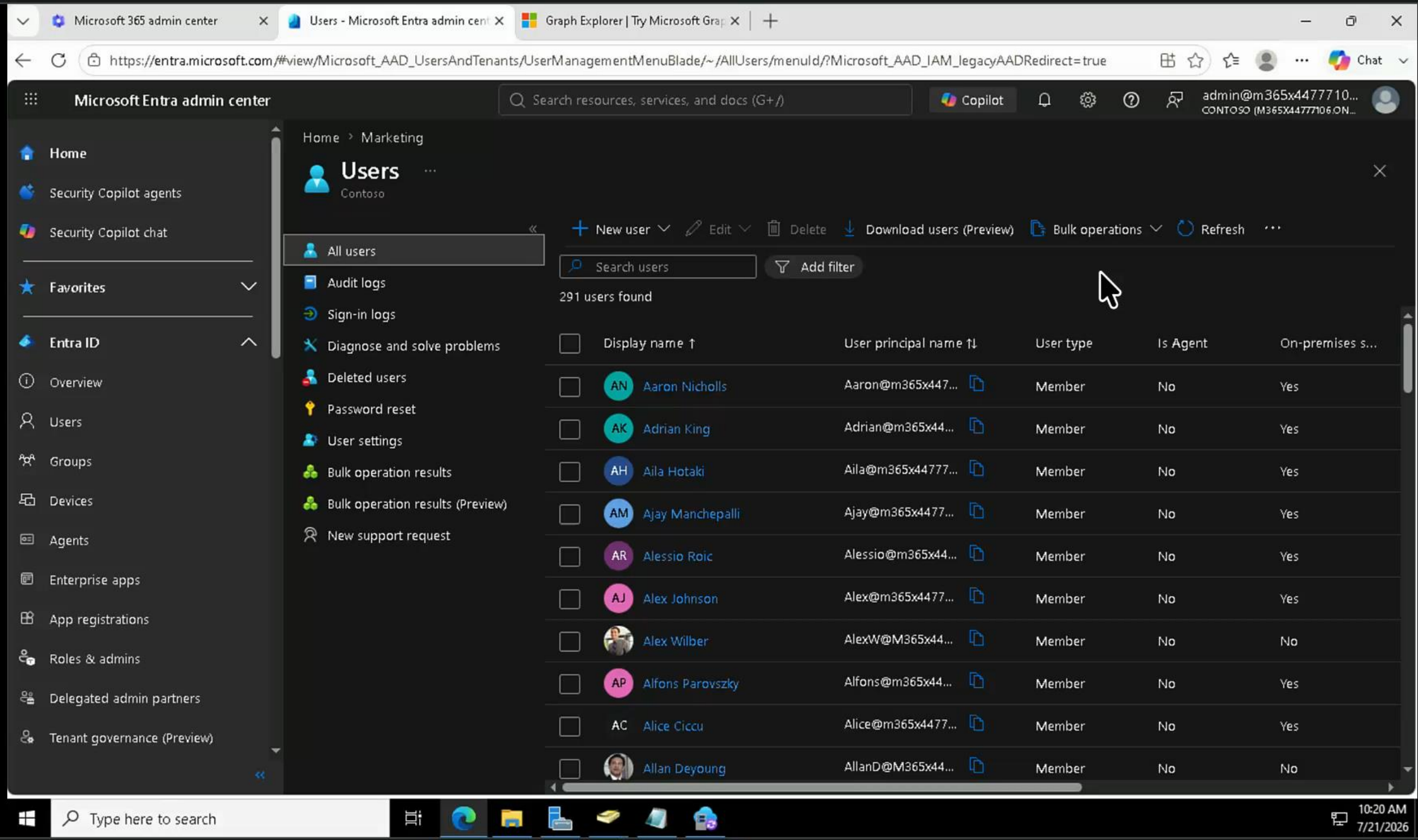
# Step 3: Flip SOA
$uri  = "https://graph.microsoft.com/v1.0/users/$($u.Id)/onPremisesSyncBehavior"
Invoke-MgGraphRequest -Method PATCH -Uri $uri -Body (@{isCloudManaged=$true}|ConvertTo-Json)

# Step 4: Confirm and inspect in admin center
(Invoke-MgGraphRequest -Method GET -Uri $uri).isCloudManaged   # True
(Get-MgUser -UserId $u.Id -Property OnPremisesSyncEnabled).OnPremisesSyncEnabled  # False
# ImmutableId is retained — this allows rollback if needed
```



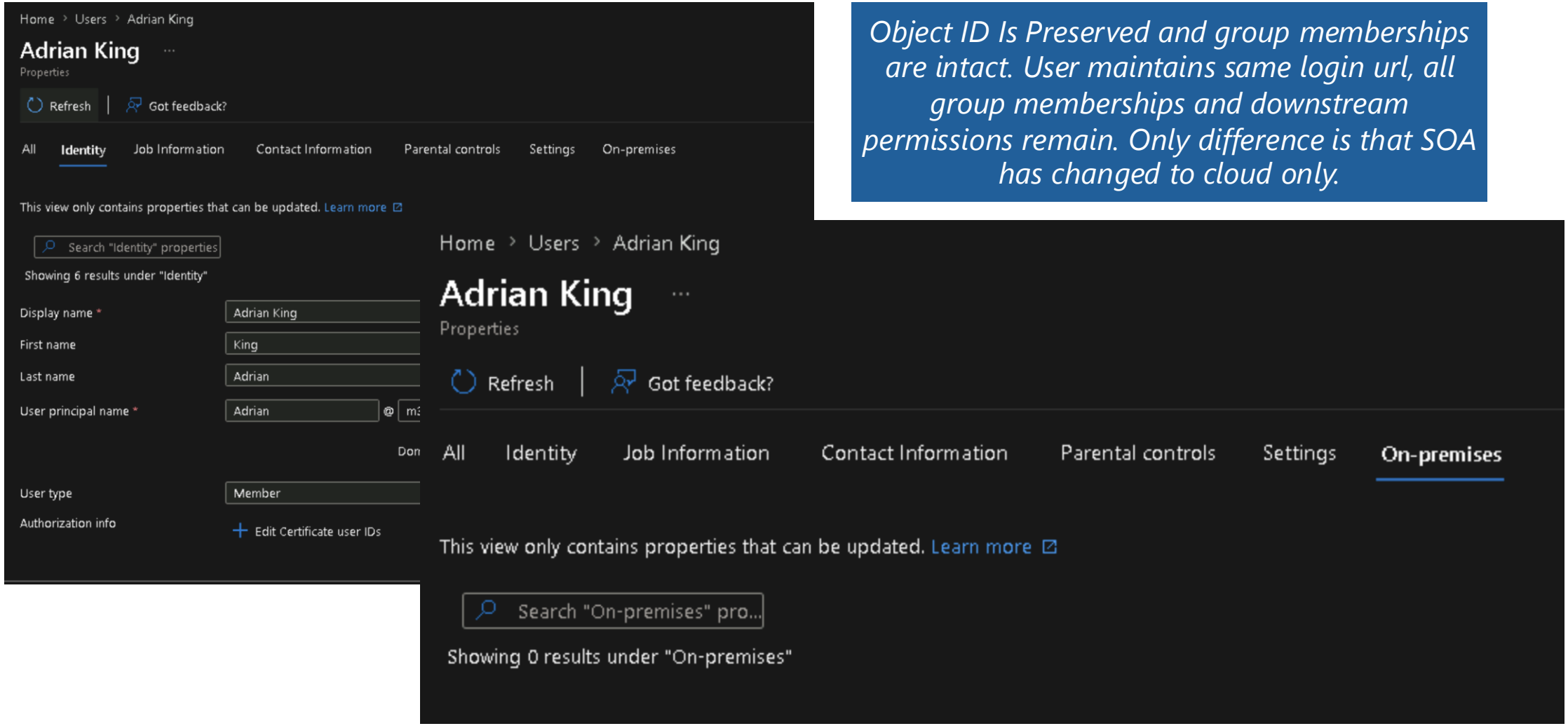
Demo

User SOA Flipping


Microsoft 365 admin center
Users - Microsoft Entra admin cent
Graph Explorer | Try Microsoft Grap
https://entra.microsoft.com/#view/Microsoft_AAD_UsersAndTenants/UserManagementMenuBlade/~/AllUsers/menuId/?Microsoft_AAD_IAM_legacyAADRedirect=true
Chat
Microsoft Entra admin center
Search resources, services, and docs (G+/)
Copilot
admin@m365x4477710...
CONTOSO (M365X44777106.ON...
Home
Security Copilot agents
Security Copilot chat
Favorites
Entra ID
Overview
Users
Groups
Devices
Agents
Enterprise apps
App registrations
Roles & admins
Delegated admin partners
Tenant governance (Preview)
Home › Marketing
Users
Contoso
All users
Audit logs
Sign-in logs
Diagnose and solve problems
Deleted users
Password reset
User settings
Bulk operation results
Bulk operation results (Preview)
New support request
New user
Edit
Delete
Download users (Preview)
Bulk operations
Refresh
Search users
Add filter
291 users found
Display name ↑
User principal name ↑↓
User type
Is Agent
On-premises s...
Aaron Nicholls Aaron@m365x447... Member No Yes
Adrian King Adrian@m365x44... Member No Yes
Aila Hotaki Aila@m365x44777... Member No Yes
Ajay Manchepalli Ajay@m365x4477... Member No Yes
Alessio Roic Alessio@m365x44... Member No Yes
Alex Johnson Alex@m365x4477... Member No Yes
Alex Wilber AlexW@M365x44... Member No No
Alfons Parovszky Alfons@m365x44... Member No Yes
Alice Ciccu Alice@m365x4477... Member No Yes
Allan Deyoung AllanD@M365x44... Member No No
Type here to search
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7/21/2026

What Happens Afterwards?

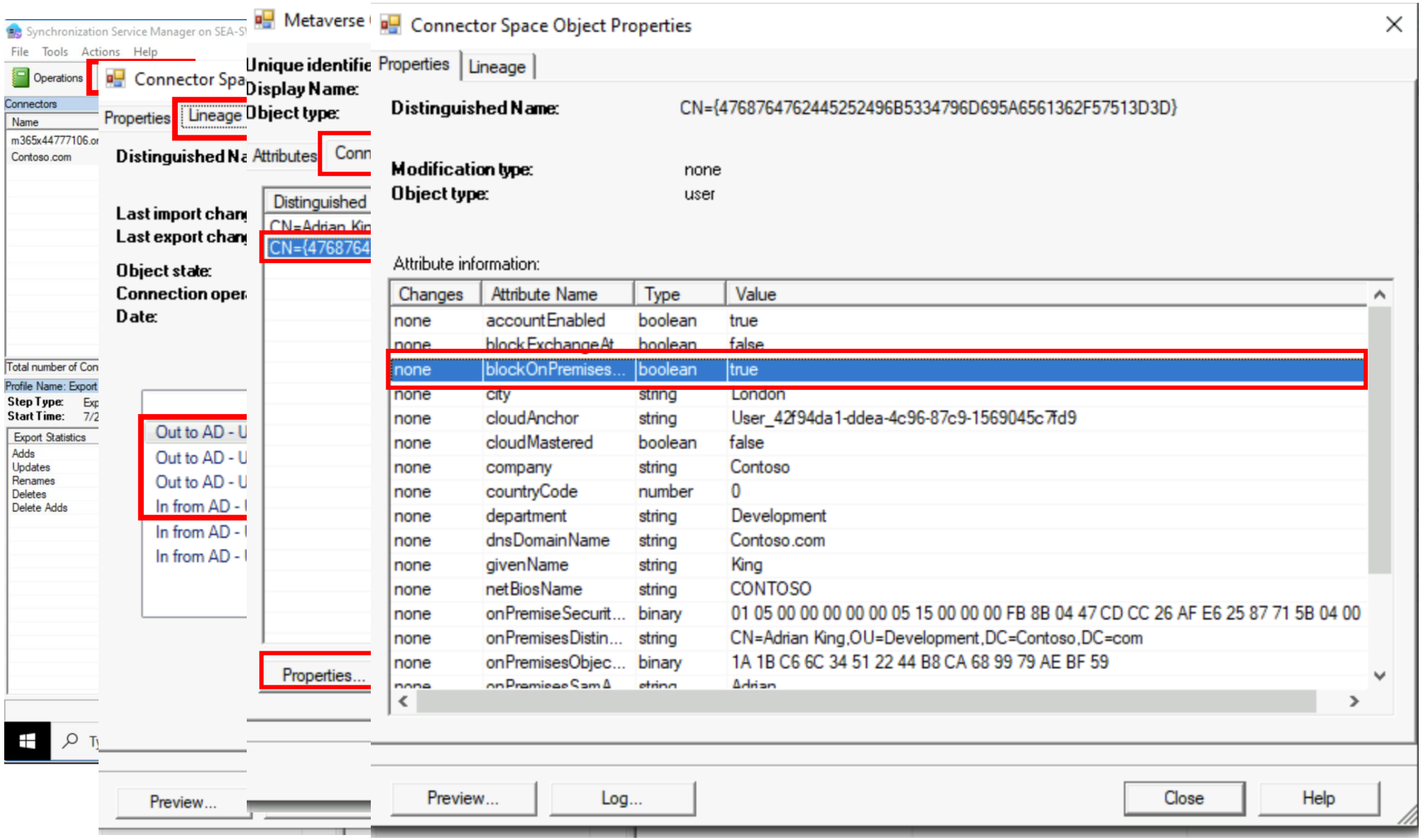
User SOA. What Happens Afterwards?



Object ID Is Preserved and group memberships are intact. User maintains same login url, all group memberships and downstream permissions remain. Only difference is that SOA has changed to cloud only.

Same object, same sign-ins, same groups. The only change — Source is now 'Cloud'.

SOA Migration Confirmation

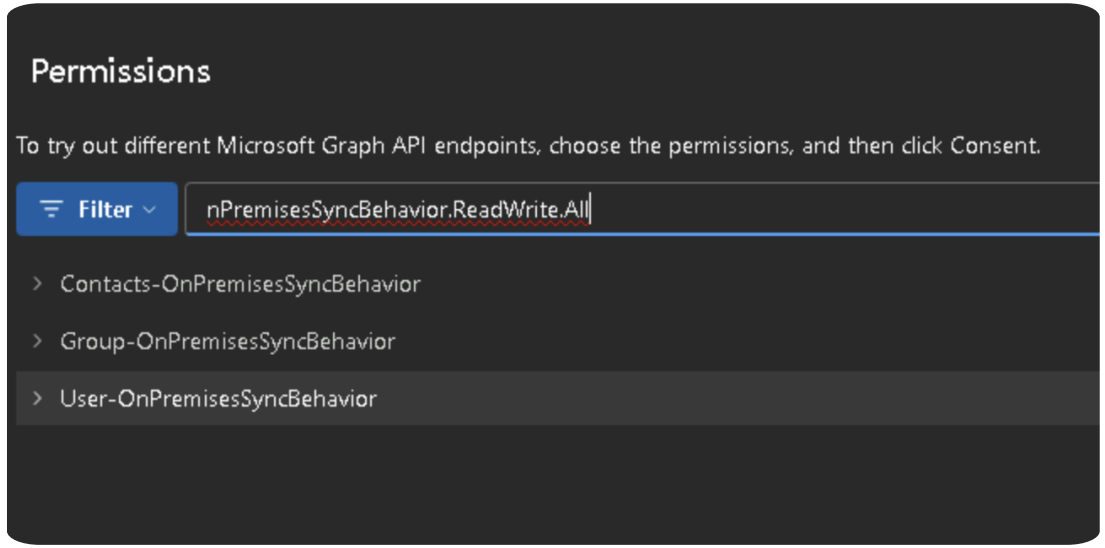




Contact Source of Authority (SOA)

Domain based Outlook Contacts migrated to Cloud only.

Changing the SOA of a Domain Based Contact



1. To transfer SOA of a Contact, the required permission is **Contacts-OnPremisesSyncBehavior.ReadWrite.All**

```
GET https://graph.microsoft.com/v1.0/contacts/{ID}/onPremisesSyncBehavior
{
   "isCloudManaged": true
}
```

1. Then Use the PATCH command

```
PATCH https://graph.microsoft.com/v1.0/contacts/{ID}/
   {
      "DisplayName": "Contact Name Updated"
   }
```



What if I need to Rollback a User?

SOA What if I Need to Rollback?

- Ensure that the users that you roll back have no cloud references.
- Remove cloud users from SOA transferred groups,
- Remove these groups from access packages before you roll back the users to AD DS
- The sync client takes over the object in the next sync cycle.



SOA Rollback Step by Step

1. Disable the **blockCloudObjectTakeoverThroughHardMatchEnabled** feature flag.
2. This flag blocks switching source of authority from cloud to on-premises by default. Issue the following request to Microsoft Graph to disable it
3. **PATCH (beta) https://graph.microsoft.com/beta/directory/onPremisesSynchronization/{id} { "features": { "blockCloudObjectTakeoverThroughHardMatchEnabled": false } }**
4. NOTE. Replace {id} with the **on-premises directory synchronization configuration ID** for your tenant
5. Next, Run the following to roll back the SOA and revert to on-premises management
6. **PATCH https://graph.microsoft.com/v1.0/users/{ID}/onPremisesSyncBehavior { "isCloudManaged": false }**
7. NOTE Replace the id with the User ID that you want to roll back.
8. **Run Start-ADSyncSyncCycle**



IMPORTANT! After the SOA Rollback

1. After the rollback is complete and Connect Sync takes over the object, re-enable the blockCloudObjectTakeoverThroughHardMatchEnabled feature flag to maintain protection against unintended cloud object takeover

```
PATCH https://graph.microsoft.com/beta/directory/onPremisesSynchronization/{id}
{
   "features": {
      "blockCloudObjectTakeoverThroughHardMatchEnabled": true
   }
}
```

2. Lastly remove any unwanted attributes user manually
   - onPremisesDistinguishedName
   - onPremisesDomainName
   - onPremisesSamAccountName
   - onPremisesSecurityIdentifier
   - onPremisesUserPrincipalName





Bulk Migration Tools?

Bulk Migration Strategy Overview

- Migration Approaches
  - Wave-based: Migrate users in scheduled batches
  - Department-based: Organize by organizational units
  - Role-based: Prioritize by job function criticality
- Critical Success Factors
  - Comprehensive testing in pilot wave
  - Roll Back Plan for Each Way
  - Communication to affected users



Bulk Migration: PowerShell Cmdlets

Essential Modules

- Microsoft.Graph.Users - Manage users and attributes
- Microsoft.Graph.Groups - Manage group membership
- Microsoft.Graph.Identity.DirectoryManagement - Control DirSync

Essential Modules

```
Install-Module Microsoft.Graph -Repository PSGallery
Connect-MgGraph -Scopes "User.ReadWrite.All",
"Group.ReadWrite.All"
```

Bulk Migration: Group Migration Strategy

Group Types Requiring Attention

- Security Groups: Migrate mailbox assignments and permissions
- Distribution Lists: Update mail routing
- Microsoft 365 Groups: Verify Teams ownership and memberships

Pre Migration Audit

```
Get-MgGroup -Filter "mail ne null" \
  -Property displayName,mail,securityEnabled | \
  Export-Csv -Path "migration_groups.csv"
```

Bulk Migration: User Migration Script



- Export Users Before Migration

```
$users = Get-MgUser -Filter "mail ne null" \
  -Property displayName,mail,userPrincipalName, \
  onPremisesImmutableId,accountEnabled
$users | Export-Csv -Path "migration_users.csv"
```

- Perform Migration for User Batch

```
$csv = Import-Csv "migration_users.csv"
foreach($user in $csv) {
  Update-MgUser -UserId $user.mail -accountEnabled $true
}
```

Bulk Migration: Validation & Testing

- Post Migration Tasks
  - Verify user sign-in to cloud applications
  - Confirm group memberships in Entra ID
  - Test conditional access policies
  - Validate MFA device registration

- Automated Validation Script

```
$report = @()
foreach($user in $csv) {
  $cloudUser = Get-MgUser -Filter "userPrincipalName eq \
    '$($user.userPrincipalName)'"
  $report += [pscustomobject]@{UPN=$user.UPN;
Status=$cloudUser.id}
}
```





Going Fully Cloud Native

Decommissioning Active Directory: Planning

Decommissioning Checklist

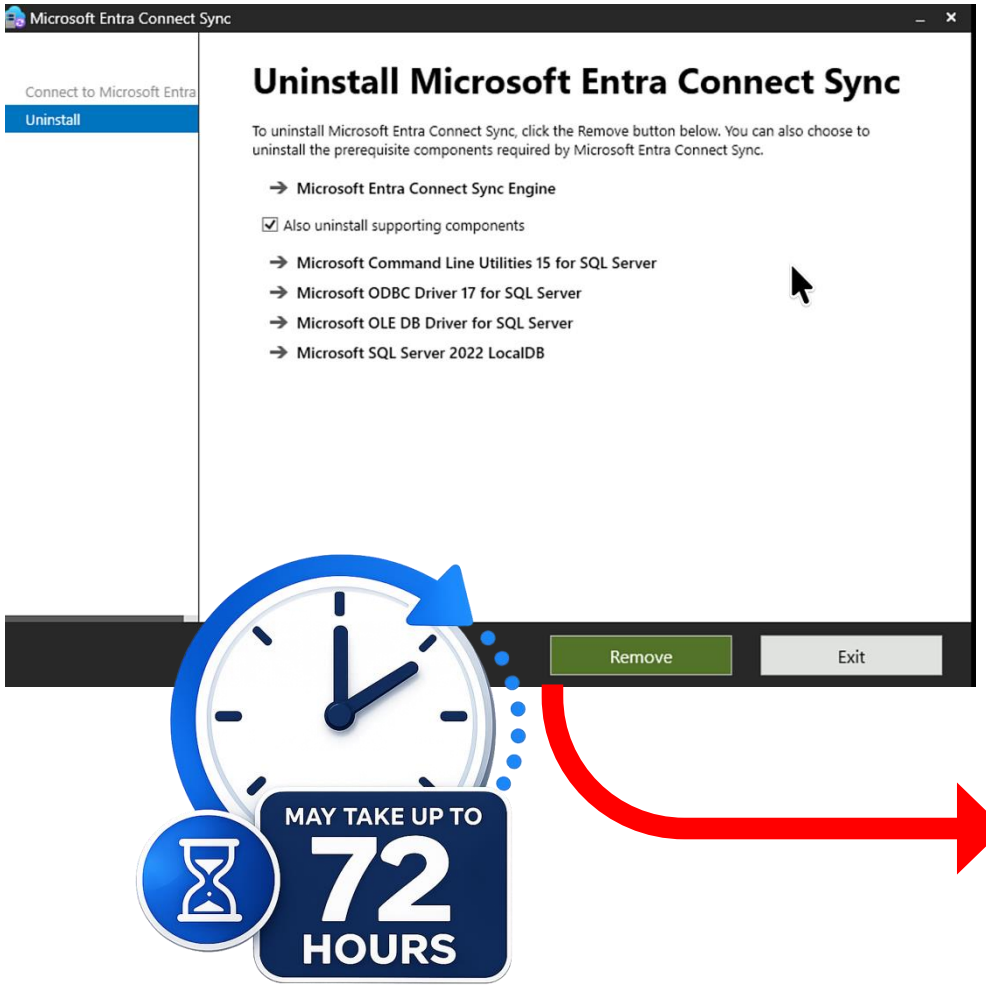
- All Devices Fully Entra Joined & Intune Managed
- All Security Groups transferred to Entra ID
- All Domain Users Fully Migrated to Entra ID (SOA Flipped)
- No on-premises authentication dependencies remaining
- Legacy system integration reviewed and updated

Extended Coexistence Period

- Recommended: 30-90 days of parallel operations
- Monitor logs for unexpected on-premises connections

Finally - Going Fully Cloud Native

Scenario: In this example the organization wants to go fully cloud native. Entra Connect removed First!



PowerShell

Copy

```powershell
# Install v1.0 Microsoft Graph PowerShell module
Install-Module Microsoft.Graph -Force

# Connect With Hybrid Identity Administrator Account
Connect-MgGraph -scopes "Organization.ReadWrite.All,Directory.ReadWrite.All"

# Verify the current status of the DirSync Type
Get-MgOrganization | Select OnPremisesSyncEnabled

# Store the Tenant ID in a variable named organizationId
$organizationId = (Get-MgOrganization).Id

# Store the False value for the DirSyncEnabled Attribute
$params = @{
  onPremisesSyncEnabled = $false
}

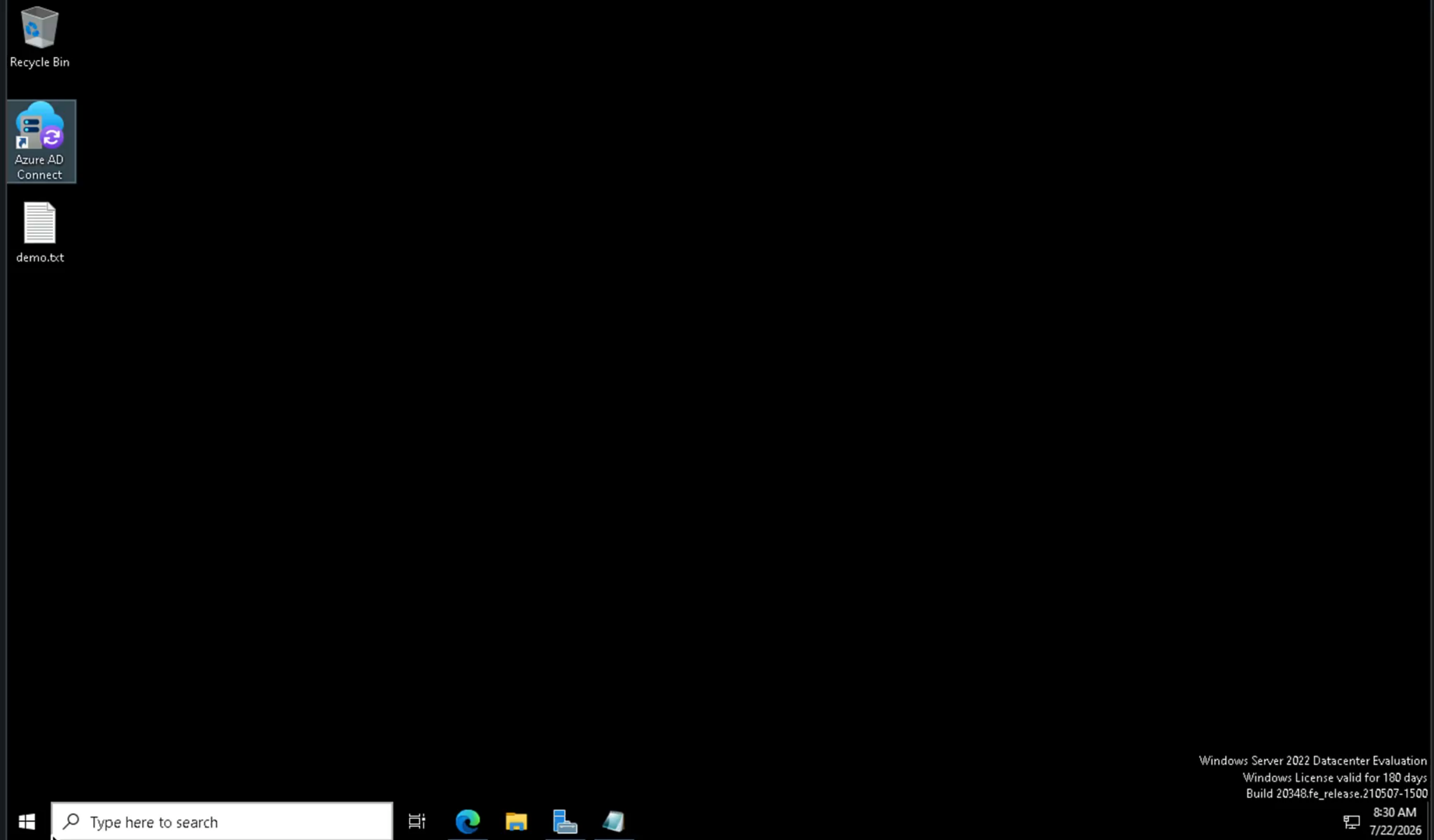
# Perform the update
Update-MgOrganization -OrganizationId $organizationId -BodyParameter $params

# Check that the command worked
Get-MgOrganization | Select OnPremisesSyncEnabled
```



Demo

Decommissioning Entra Connect Sync


Recycle Bin
Azure AD Connect
demo.txt
Windows Server 2022 Datacenter Evaluation
Windows License valid for 180 days
Build 20348.fe_release.210507-1500
Type here to search
8:30 AM
7/22/2026

Next: Decommissioning AD: Decommission Servers

- Disable AD Domain Services



```
# PowerShell on Domain Controller:
Uninstall-ADDSDomainController -Force
# Or use Server Manager > Manage > Remove Roles and Features
```

- Pre-Decommission Validation
  - No applications authenticating against AD
  - All group policies migrated or replaced with Intune
  - File share permissions updated for cloud identities
  - 30-day hold period for emergency recovery

- Export Data for Compliance

```
# Export all AD objects before removal:
Get-ADUser -Filter * -Properties * | Export-Csv \
  "ad_users_backup.csv"
Get-ADGroup -Filter * -Properties * | Export-Csv \
  "ad_groups_backup.csv"
Get-ADComputer -Filter * -Properties * | Export-Csv \
  "ad_computers_backup.csv"
```

- Compliance Archival
  - Store exports in secure, long-term archive
  - Document removal dates and approvals

Windows Server Removal



- Pre Removal Assessment
  - Verify no domain-joined workstations depend on DC
  - Ensure all Applications are decommissioned & Migrated to Entra / M365
  - Migrate All Shares, File Servers to Cloud or Stand Alone
  - Migrate DHCP and DNS to cloud services or Windows Server offline
  - Update all VPN and remote access to use Entra ID
- DNS Migration Strategy
  - Migrate from on-premises DNS to Azure DNS
  - Update name servers in domain registrar
  - Ensure all cloud based DNS Records are updated & correct
- Backup & Rollback
  - Ensure you have full backups and able to Roll Back if Problems Occur
  - Keep Good documentation


Mission Accomplished!

Conclusions

- Understand SOA Migration
- Understand the Function of Microsoft Graph
- Understand on What SOA Records are & ho they work
- Change Group SOA Attributes to Cloud Native
- Change User SOA Attributes to Cloud Native
- Change Contact SOA Attributes to Cloud Native
- What Happens Afterwards
- How to Rollback an Object
- Flick the Switch to Cloud Native

Thanks for Attending!